INCH-POUND

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MILITARY SPECIFICATION

BOOTS, INTERMEDIATE COLD/WET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers black waterproof, mildew resistant, insulated combat boots with a closed loop speed lace system and a welt sole for wear in intermediate cold/wet climates.

1.2 Classification. The boots shall be of one type in the following sizes and widths as specified (see 6.2).

Schedule of sizes

Width 1/	3	3-1/2	4	4-1/2	5	5-1/2	6	6-1/2	7	7-1/2	8	8-1/2
N	X	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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Schedule of sizes (cont'd)

Width 1/	9	9-1/2	10	10-1/2	11	11-1/2	12	12-1/2	13	13-1/2	14
N	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X

1/ N = B, R = D, W = EE, and XW = EEE

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Wrapping
A-A-1249	-	Paper, Wrapping, Tissue
V-L-61	-	Laces, Nylon
V-T-295	-	Thread, Nylon
KK-W-231	-	Welting, Leather, Shoe
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet-Stock (Container Grade) and Cut Shapes

MILITARY

MIL-L-3122	-	Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored
MIL-T-5624	-	Turbine Fuel, Aviation Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-A-8243	-	Anti-icing and Deicing-Defrosting Fluids
MIL-L-10867	-	Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-41814	-	Counter, Footwear

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-311 - Leather, Methods of Sampling and Testing
FED-STD-601 - Rubber; Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1635 - Speed Lace Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

Gray Scale for Color Change

AATCC 8 - Colorfastness to Crocking
AATCC 15 - Colorfastness to Perspiration

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AATCC 107 - Colorfastness to Water

(Copies should be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 573	-	Rubber Deterioration in an Air Oven
D 624	-	Rubber Property - Tear Resistance
D 1565	-	Specification for Flexible Cellular Materials - Vinyl Chloride Polymers and Copolymers (Open Cell Foam)
D 1630	-	Rubber Property - Abrasion Resistance (NBS Abraded)
D 2240	-	Rubber Property - Durometer Hardness
D 3574	-	Method of Testing Flexible Cellular Materials - Slab, Bonded and Molded Urethane Foams
E 96	-	Water Vapor Transmission of Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Leather.

3.3.1.1 Upper leather vamps, quarters, eyelet stays, and backstays. The upper leather shall conform to treatment A and treatment B of MIL-L-3122 with a mellow temper, except that the leather shall be full grain only. The eyelet stays shall be thinner pliable portions of leather split to a thickness of not less than 2-1/2 ounces nor more than 3-1/2 ounces. The

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leather shall be drum dyed black with a grain surface finish that when scuffed, may be restored to uniformity by simple brushing or application of commercial polish containing silicone. The leather shall be finished so that it has an adequate moisture vapor transmission rate for breathable footwear. The leather shall conform to style Thule J74000 (see 6.5.1.1); style Boulder HH13339 (see 6.5.1.2) or equal (see 6.7).

3.3.1.2 Tongues and collars. The tongues and collars shall be gusset leather conforming to treatment C of MIL-L-10867. The color shall be black. At least 80 percent of the eyelet stay specimens tested shall have a slot tear strength of not less than 30 pounds when tested in accordance with MIL-L-3122.

3.3.2 Insoles. The insole shall be cut from Texorist material (see 6.5.2). The Texorist shall contain 0.25 to 1.00 percent copper-8-quinolinolate uniformly distributed throughout the insole. The Texorist insole shall be 6 irons or 0.126 ± 0.003 inch thick. The insole shall be cut with the heel-to-toe direction across the machine direction of the Texorist. The Texorist insole shall have a stuck-on rib applied to the printed side of the material.

3.3.3 Fabrics.

3.3.3.1 Bootie lining. The bootie lining shall be constructed from no more than three pattern pieces cut from a waterproofed five-layer laminated quilted package material (see 6.5.3).

3.3.3.1.1 Five-layer laminate. The five-layer laminate lining consists of a three-layer quilted package, a plastic film, and a tricot knit cloth conforming to the following requirements.

3.3.3.1.1.1 Quilted package. The three-layer quilted package (see 6.5.4) shall consist of the following:

a. Top layer. The top layer shall consist of a thermal bonded Nylon 6,6/Nylon 6. The weight shall be 4.3 ± 0.7 ounces per square yard. The thickness shall be 0.028 ± 0.008 inch.

b. Middle layer. The middle layer shall be 65 percent polyolefin and 35 percent polyester. The weight shall be 210 g/m^2. The thickness shall be 0.55 inch. The R-value shall be 2.2.

c. Bottom layer. The bottom layer shall consist of a thermal bonded Nylon 6,6. The weight shall be 0.5 ± 0.1 ounces per square yard.

3.3.3.1.1.2 Plastic film. The plastic film shall be polytetrafluoroethylene, microporous. The weight shall be 0.5 ± 0.02 ounces per square yard.

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3.3.3.1.1.3 Tricot knit. The tricot knit cloth shall be nylon. The weight shall be 0.8 ± 0.10 ounces per square yard. The color shall be natural.

3.3.3.2 Adhesive. The adhesive for laminating the layers together shall be such that the laminated cloth will meet the requirements of this specification. The adhesive shall have no adverse health hazard when used as intended.

3.3.3.3 Laminated cloth. The quilted package specified above shall be laminated to one side of the plastic film and the tricot knit cloth shall be laminated to the other side of the plastic film. The face side of the laminated cloth shall be the side with the quilted package. The laminated cloth shall conform to the requirements in table I when tested as specified in 4.4.1.

TABLE I. Physical requirements

Characteristics	Minimum	Maximum
Weight, oz/sq yd	11.00	15.00
Thickness (inches)	0.070	0.170
Thermal shrinkage:		
Machine (%):		
Sample mean	0.88	-2.42
Lot mean	0.86	-2.33
Thermal shrinkage:		
Transverse (%):		
Sample mean	1.16	-2.35
Lot mean	1.15	-2.41
Moisture vapor transmission rate (g/m^2/24 hrs):		
Procedure B	640	—
Procedure BW	2770	—
Water permeability		
Initial 3 psi, 5 minutes	no leakage	
After aircraft fluid	no leakage	

3.3.3.3.1 Color. The color of the face side of the laminated cloth shall be a solid Nicotine and the color of the back side, or tricot knit side, shall be natural. Both sides of the cloth shall match the respective sides of the standard sample.

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3.3.3.3.2 Matching. The color shall match the standard sample (see 6.4) when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500 ± 200K with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.3.3.3.3 Colorfastness. The face side of the laminated cloth shall meet the colorfastness requirements specified below, when tested in accordance with 4.4.1. Test one specimen per sample. The light equipment shall be Macbeth Model #SPL-75, or equal (see 6.7).

Characteristic	Colorfastness requirements
Fastness to water: 1/	
Color change 2/	4 minimum
Staining 3/	4 minimum
Fastness to perspiration: 1/	
Color change 2/	4 minimum
Staining 3/	3-4 minimum
Fastness to crocking:	
Dry 2/	3-4 minimum
Wet 2/	3-4 minimum

1/ Remove the top layer from the laminated cloth specimens before testing.

2/ Evaluate color change using the AATCC Gray Scale for Color Change.

3/ Evaluate staining using AATCC Chromatic Transference Scale. Colorfastness requirements are based on the lowest rating given any fiber on multifiber test fabric number 10.

3.3.4 Tapes.

3.3.4.1 Seam sealing tape for bootie. The tape for sealing all designated seams and stitching in the bootie shall be cut in 7/8 + 1/16, -0 inch wide strips from natural colored laminated cloth fabricated in layers as follows (see 6.5.5.1):

a. A nylon tricot knit cloth weighing 0.80 ± 0.10 ounces per square yard.

b. The middle layer shall conform to the plastic film specified in 3.3.3.1.1.2.

c. Adhesive, a minimum of 5.4 mils in thickness.

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3.3.4.2 Seam sealing tape for heel and counter area of bootie. The tape used to apply to the heel and counter area shall be a flexible two-layer film tape (see 6.5.5.2). It shall be designed to eliminate the tape blocking during application, and to impart strength and abrasion resistance to the sealed seams. It shall combine a low melting polyester adhesive with a very flexible polyurethane carrier layer. The tape shall be fabricated in layers as follows and shall meet the following requirements when tested as specified in 4.4.1.

a. Adhesive layer. The adhesive layer shall be polyester. The thickness shall be 0.004 inch. The color shall be cloudy, translucent with a slight amber cast.

b. Carrier layer. The carrier layer shall be an ester-based polyurethane elastomer. The thickness shall be 0.002 inch. The color shall be hazy with an amber cast.

3.3.5 Collar foam. The foam for the collar shall be 3/8 ± 1/32 inch polyurethane open-cell foam having a density of 2 pounds per cubic foot.

3.3.6 Thermal quarter insulation. The thermal insulation shall be a light-weight, warm, moisture resistant, breathable 100 percent polyolefin. The thickness shall be 0.3 inch. The weight shall be 12.4 ounces per square yard.

3.3.7 Stuck on rib. The stuck on rib shall go completely around the insole, starting and stopping on the inside shank area.

3.3.7.1 Rib construction. The ribbing fabric used in the construction of the rib shall be unbleached cotton or cotton-synthetic fabric conforming to the requirements of table II when tested as specified in 4.4.1. The fabric shall be coated on one side with a rubber-base adhesive and bonded to the printed side of the insole. The fabric or fiberboard shall cover 5/32 -2/32, +1/32 inch of the peripheral edge and extend over the top of the vertical rib, or the fabric may extend from the inside top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The preformed rib shall be firmly and smoothly fitted and bonded to the flesh side of the insole as specified above to provide for the required edge extension on the finished boot.

TABLE II. Ribbing fabric requirements

Characteristic	Requirement, minimum
Weight	8.0 oz/sq yd
Yarns per unit length:	
Warp	52 yarns per inch
Filling	30 yarns per inch

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TABLE II. Ribbing fabric requirements (cont'd)

Characteristic	Requirement, minimum
Breaking strength:	
Warp	105 pounds
Filling	105 pounds

3.3.7.2 Rib strength. The physical requirements for the rib strength shall conform to table III when tested as specified in 4.4.1.

TABLE III. Rib strength requirements

Characteristic	Minimum 1/	Average
Shear strength	70 pounds	75 pounds
Stitch strength	20 pounds	30 pounds

1/ No single determination (sample unit) shall fall below the minimum value specified and the average of all specimens (lot average) shall be not less than the average specified.

3.3.8 Box toe. The box toe shall be of virgin polyethylene, natural and unfilled (see 6.5.6). The specific gravity shall be 0.942 and the melt flow shall be 11.5. The thickness shall be an average of 0.045 to 0.055 inch. As an alternate, the box toe material shall be a laminate of 0.020 ± 0.002 inch virgin ionomer resin (surlyn) sandwiched between two layers of nonwoven polyester saturated with polystyrene. The total thickness shall be 0.056 ± 0.005 inch. The box toes shall be skived 3/8 ± 1/8 inch wide at the breast.

3.3.8.1 Box toe sizes. Box toes shall be made in seven sizes to cover the range of boot sizes and widths specified in table IV. The box toes shall be fitted in accordance with table IV. When the alternate box toe is used, the box toes shall be made in a sufficient number of sizes to cover boot sizes 3 through 14 in the necessary widths.

TABLE IV. Box toe fitting schedule

Box toe size	Boot width and sizes			
	N	R	W	XW
2	3 - 3-1/2	3	—	—
4	4 - 5-1/2	3 - 5-1/2	3 - 4-1/2	3 - 4
6	6 - 7-1/2	5-1/2 - 7	5 - 6-1/2	4-1/2 - 6

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TABLE IV. Box toe fitting schedule (cont'd)

Box toe size	Boot width and sizes			
	N	R	W	XW
8	8 - 9-1/2	7-1/2 - 9	7 - 8-1/2	6-1/2 - 8
10	10 - 11	9-1/2 - 10	9 - 10-1/2	8-1/2 - 10
12	12 - 13-1/2	11-1/2 - 13	11 - 12-1/2	10-1/2 - 12
14	14	13-1/2 - 14	13 - 14	12-1/2 - 14

3.3.9 Counters. The counters shall conform to the requirements of MIL-C-41814. The counters shall be made in a sufficient number of sizes to cover boot sizes 3 through 14 and in the necessary widths. Counters shall conform to the specific last used.

TABLE V. Counter fitting schedule

Counter size	Boot width and sizes			
	N	R	W	XW
000	3 - 4-1/2	--	--	--
00	5 - 6-1/2	3 - 4-1/2	--	--
0	7 - 8-1/2	5 - 6-1/2	3 - 4-1/2	--
1	9 - 10-1/2	7 - 8-1/2	5 - 6-1/2	3 - 4-1/2
2	11 - 12-1/2	9 - 10-1/2	7 - 8-1/2	5 - 6-1/2
3	13 - 14	11 - 12-1/2	9 - 10-1/2	7 - 8-1/2
4	--	13 - 14	11 - 12-1/2	9 - 10-1/2
5	--	--	13 - 14	11 - 12
6	--	--	--	13 - 14

3.3.10 Welting. The welting shall conform to the requirements of KK-W-231. As an alternate, the vinyl welting shall be of flat top design, bright black in color and may be formed with or without storm welt of medium size bead. The welting shall be of sufficient width to ensure that the required extensions on the finished boots are met.

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3.3.11 Fiberglass shank. The fiberglass shank shall consist of glass fibers impregnated with vinyl ester resins encased in a plastic sleeve or tube (see 6.5.7). The width of the uncured encased fiberglass resins shall be 5/8 ± 1/32 inch. The shank shall be cured and bonded to the insole and conform to the shape of the last. When fully cured, the shank (glass and resin area) shall be 5/8 ± 1/16 inch in width. The forward end of the shank shall be tapered for a minimum of 3/8 inch. Testing shall be as specified in 4.4.1.

3.3.11.1 Shank fitting sizes. The shank fitting shall conform to the schedule as specified in table VI.

TABLE VI. Shank lengths (± 1/8 inch)

Shank size	5	5-1/4	5-1/2	5-3/4	6	6-1/4
Boot sizes (all widths)	3 - 4	4-1/2 - 6	6-1/2 - 8	8-1/2 - 10	10-1/2 - 12	12-1/2 - 14

3.3.11.2 Shankboard. The shankboard shall be cut from Bonfoam 300MD (see 6.5.8) or equal (see 6.7).

3.3.12 Metal.

3.3.12.1 Staples. A staple lasting shoe wire described as follows shall be used:

Size	022A
Finish	Brassy
Package	70 pound fiber spools
Tensile	130,000 to 160,000 psi

Chemical analysis:

Carbon	0.15 to 0.20
Manganese	0.60 to 0.90
Phosphorus	0.04 maximum
Sulphur	0.05 maximum
AISI or SAE Number	C-1018

3.3.12.2 Eyelets. The eyelets shall be gloss finish nylon coated brass, X-long, conforming to size AA and have a dark oxide finish.

3.3.12.3 Speed lace assembly. The speed lace assembly shall be a two-piece construction consisting of gloss finish nylon coated brass speed

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lacer and nylon coated alternate full tubular rivet in accordance with Drawing 2-1-1635. The color shall be black.

3.3.13 Bottom filler.

3.3.13.1 Cold process bottom filler. The cold process bottom filler shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in the proportion of two parts cork to one part binder by volume. The cork granules shall be free from bark. When the spread filler is dry and set, it shall consist of four parts cork to one part binder. It shall not soften at less than 150°F and shall be water resistant and flexible. As an alternate, when applied by semi-automatic equipment or method, the cold process bottom filler shall consist of a mixture of ground cork and a suitable binder in the proportion by weight of one part cork to five parts binder. Upon loss of solvent, it shall consist by weight of one part cork to 3-3/4 parts binder. The binder shall be water insoluble and have a softening point of at least 125°F.

3.3.14 Rubber compounds.

3.3.14.1 Midsole. The midsole shall be a 6 iron (1/8 inch) black neoprene.

3.3.14.2 Outersole and heel. The outersole and heel shall be a black non-marking rubber conforming to Vibram 134AR (see 6.5.9) or equal (see 6.7).

3.3.14.3 Rubber compound requirements. The requirements for the rubber are based on testing done on test slabs and not actual soles and according to ASTM procedures. There may be some deviation on actual outersoles or midsoles based on style thickness and curing parameters. The requirements shall be as specified below.

Style	134AR (outersole)	6 iron midsole
Compound	344	330
NBS abrasion	150 min.	N/A
Hardness (unaged)	70-80A	N/A
Trouser tear	65 lb. min.	N/A
Nail tear (dry)	320 lb. min.	180 lb. min.
Nail tear (fuel B)	225 lb. min.	50 lb. min.
C tear	308 lb. min.	N/A

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Style	134AR (outersole)	6 iron midsole
Ross Flex, Oil #3	250% max., 100,000 flexes	N/A
Volume swell (fuel B)	30% max.	30% max.

3.3.14.4 Bond strength. The outersole to midsole bond strength shall be not less than 100 pounds when tested as specified in 4.4.4.

3.3.15 Thread.

3.3.15.1 Thread, upper fitting. The thread for all upper fitting operations shall conform to type III, class B (water repellent), size E of V-T-295. The color shall be black.

3.3.15.2 Thread, inseaming. The thread for all inseaming operations shall conform to size E of V-T-295, and shall be water repellent. The color shall be natural.

3.3.15.3 Thread, sole stitching. The thread for sole stitching shall be polyester. The color of the running thread shall be black and the color of the bobbin thread shall be natural. The bobbin thread shall be coated with wax.

3.3.16 Wax, inseam sewing and sole stitching. The wax shall be white or golden and shall be a permanently plasticized resin. The wax, in the temperature range of normal machine use, shall thoroughly wax the threads described in 3.3.15.2 and 3.3.15.3.

3.3.17 Adhesives. The adhesives used for bonding the various leather and fabric components of the boots shall be of the following types:

a. Natural rubber latex
b. Synthetic rubber latex (including chloroprene)
c. Natural rubber solvent cement
d. Synthetic rubber solvent cement
e. Synthetic resin cement
f. Polyurethane rubber solvent cement
g. Thermoplastic hot melt

3.3.18 Finishing materials for leather uppers. Finishing materials used shall be compatible with the leather's silicone waterproofing agents and cause no adverse effect on water resistance of the leather or boots.

3.3.18.1 Repairers. Repairers shall be liquid spray, crayon, or paste type applied by hand. The color shall match the upper leather and have sufficient coverage to correct minor surface imperfections of the leather.

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3.3.18.2 Fillers. Fillers, where used, shall be capable of providing a suitable foundation for the application of the top finish and shall be applied by sponge or spray method.

3.3.18.3 Top finish. The top finish shall be applied by either the sponge or spray method and shall be of synthetic resin and wax base to even out the tannery finish but not to cover or change its natural appearance. It shall finish dry to a high gloss.

3.3.19 Instruction tag. The instruction tag, to be tied to the finished boots, shall be printed on tag stock that is 5-1/4 by 6-1/2 inches and folded in the middle to form 5-1/4 by 3-1/4 inch printed pages in book form. A punch hole shall be made in the top near the folded edge for insertion of the lace. The instruction tag shall be printed using the data specified on figure 1.

3.3.20 Molded cushion insole. The molded cushion insole shall be a medium brown microcellular urethane foam, with an integral skin laminated to a thermo-formed cross-link polyethylene heel cup/arch support. The polyurethane cover shall have a thickness of 0.110 ± 10 inch, and shall include an active organic anti-microbial agent (see 6.5.11). The top cover shall be perforated. The polyurethane foam shall conform to the following properties:

Properties	Requirements
Density	17 + 2, -1 pounds per cubic foot
Tensile strength	115 pounds per square inch (min.)
Compression set	10 percent (max.)
Compression set after hydrolysis	10 percent (max.)

The size and width of each insole shall be legibly stamped on the bottom (foam side) of the insert in the shank area in numbers and letters not less than 3/8 inch in height. The size of the cushion insole shall be that needed to cover the last bottom of the whole boot and shall be used in such size boot and the next larger half size (size 9R insole to be used in 9R and 9-1/2R). There shall be three widths only in Regular, Narrow, and Wide. The Narrow shall be used in N and the Wide in XW. The insole shall be cut in such a manner that the full contoured portion of the heel area and arch support will be included. The total thickness of the insole shall be 0.130 to 0.155 inch forward of the instep arch and 0.190 to 0.215 inch elsewhere.

3.3.21 Laces. The laces shall conform to type III, class 1 of V-L-61. The color of the laces shall be black. The length of each lace shall be a minimum of 72 inches for sizes 3 through 6-1/2, 87 inches for sizes 7 through 11, and 97 inches for sizes 11-1/2 through 14.

3.4 Design. The design shall be for a full lace closure with eyelet row edge of the quarters, grain-out leather upper, plain toe vamp, backstay, gusset tongue, box toe, and three pattern piece bootie. The boot shall have

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a neoprene midsole and a full length rubber base outsole with a polyurethane midsole.

3.5 Pattern and dies, number 9218. The height of the finished boot, measured upward on outside from bottom at breast of heel to top of boot, shall be 11 inches on size 9 and shall graduate up or down between whole sizes approximately 1/8 inch as indicated by patterns. A standard set of paper patterns will be furnished by the Government as a basis from which the contractor's dies or patterns shall be made. The patterns and contractor's cutting dies and patterns shall consist of the component parts as specified in table VII. The Government patterns shall not be altered in any way and shall be followed, except for lasting allowance, which shall be determined by the contractor. The contractor shall furnish insole paper patterns. When approved by the procuring activity, the bootie patterns may be altered to properly fit the last (see 3.6).

TABLE VII. Contractor's dies and patterns, number 9218

Component parts	Widths	Sizes
Vamps	4 widths	Whole and half sizes
Quarter	4 widths	Whole and half sizes
Tongue	4 widths	3 - 6-1/2, 7 - 8-1/2, 9 - 14, EEE 3 - 6-1/2, 7 - 9, 9-1/2 - 14, EE 3 - 6-1/2, 7 - 9-1/2, 10 - 14, D 3 - 6-1/2, 7 - 9-1/2, 10 - 14, B
Eyelet stay	4 widths	3 - 6-1/2, 7 - 8-1/2, 9 - 14, EEE 3 - 6-1/2, 7 - 9, 9-1/2 - 14, EE 3 - 6-1/2, 7 - 9-1/2, 10 - 14, D 3 - 6-1/2, 7 - 9-1/2, 10 - 14, B
Collar	1 width	3 - 6-1/2, 7 - 9-1/2, 10 - 11-1/2, 12 - 13-1/2, 14, B-D-EE-EEE
Backstay	4 widths	3 - 6-1/2, 7 - 8-1/2, 9 - 14, EEE 3 - 6-1/2, 7 - 9, 9-1/2 -14, EE 3 - 6-1/2, 7 - 9-1/2, 10 - 14, D 3 - 6-1/2, 7 - 9-1/2, 10 - 14, B
Quarter filler	4 widths	Whole sizes
Collar foam	1 width	3 - 6-1/2, 7 - 9-1/2, 10 - 11-1/2, 12 - 13-1/2, 14, B-D-EE-EEE
Inside bootie	4 widths	Whole and half sizes

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TABLE VII. Contractor's dies and patterns, number 9218 (cont'd)

Component parts	Widths	Sizes
Outside bootie	4 widths	Whole and half sizes
Bootie bottom	4 widths	Whole and half sizes
Insoles	4 widths	Whole and half sizes
Shankboard	N/A	6 sizes

Machine Marking Equipment

9218 Vamp marker line for tongue location and quarter front location:

3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
	7 - 9-1/2, D	10 - 14, D
	7 - 9-1/2, B	10 - 14, B

9218 Quarter marker line for top of vamp location:

3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
	7 - 9-1/2, D	10 - 14, D
	7 - 9-1/2, B	10 - 14, B

9218 Back tongue marker line for centering top of bootie lining and 6 stitches through front of tongue to hold bootie on center:

3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
	7 - 9-1/2, D	10 - 14, D
	7 - 9-1/2, B	10 - 14, B

9218 Front tongue marker line for stitching single needle across front of tongue through the bootie lining:

3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
	7 - 9-1/2, D	10 - 14, D
	7 - 9-1/2, B	10 - 14, B

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TABLE VII. Contractor's dies and patterns, number 9218 (cont'd)

9218 Eyelet stay marker line for boxstitching bottom of eyelet stay and tunnel hook location stab in cutting dies:		
3 - 6-1/2, D-EE-EEE	7 - 8-1/2, EEE	9 - 14, EEE
3 - 6-1/2, B	7 - 9, EE	9-1/2 - 14, EE
	7 - 9-1/2, D	10 - 14, D
	7 - 9-1/2, B	10 - 14, B

Cutting instructions	Lasting instructions
Cut N up 1/2 on B	Last up 1/2 on M2673B-1 B
Cut R up 1/2 on D	Last up 1/2 on M2673B-1 D
Cut W up 1/2 on EE	Last up 1/2 on M2673B-1 EE
Cut XW up 1/2 on EEE	Last up 1/2 on M2673B-1 EEE

3.6 Lasts. The boots shall be made on model number M2673B-1 last (see 6.5.12.1), 701 last (see 6.5.12.2) or equal (see 6.7). A full set of patterns consisting of vamp, quarter, tongue, eyelet stay, backstay, bootie inside and outside, also bootie bottom and insoles will be loaned to the contractor by the Government.

3.7 Construction.

3.7.1 Cutting uppers. The uppers shall be cut from grain out leather. The leather shall conform to the thickness specified in table VIII. Except for the eyelet stay, none of the upper parts shall be split or shaved. No parts cut off stretch shall be accepted. Vamps shall be cut from the bend area of the side leather.

TABLE VIII. Thickness of leather uppers

Part	Thickness (ounces)	
	Minimum	Maximum
Vamp	5	6
Quarter	4-1/2	6
Eyelet stay	2-1/2	3-1/2
Gusset	2-1/2	3-1/2

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3.7.2 Skiving. The skiving of upper leather parts shall conform to the requirements specified in table IX.

TABLE IX. Skiving requirements

Part	Side	Location	Scarf width (inches)	Scarf edge thickness in millimeters (+ .03 mm)
Quarter	Flesh	Vamp location	3/4 ± 1/8	1.5
Quarter	Flesh	Eyestay location	3/4 ± 1/8	1.2
Vamp	Flesh	Lasting margin	3/4 ± 1/8	1.5
Vamp	Flesh	Rear top line	1/2 ± 1/8	1.3
Backstay	Flesh	Lasting margin	3/4 ± 1/8	1.5
Eyelet stay	Flesh	Top line	3/4 ± 1/8	1.1

3.7.3 Marking, permanent identification. The inside quarter of each boot shall be marked on the grain side with the correct size and width, the contractor's identification symbol, and the month and year (expressed numerically) of the date of contract. This marking shall be impressed into the grain side of the leather in such a manner as to be permanently visible without cutting through at any point, and shall be placed approximately in the center of the inside quarter and between collar stitching. Figures shall be Arabic and letters shall be Gothic. The figures and letters shall be a minimum of 9/32 inch and a maximum of 3/8 inch in height. The contractor's symbol shall be in a block as shown by the following example: 10W /AB/ 6-92.

3.7.4 Tongue construction. The seam in the lower front of the tongue is lap seamed with the upper part lapped over to cover the stab in tongue. This is stitched on a single needle machine and backtacked on both ends.

3.7.5 Bootie lining construction. The lining material shall be cut with the diamond quilting stitching running long from heel to toe. The lining consists of three cut pieces: outside quarter, inside quarter, and bottom. The thread for the construction of the bootie shall be white with a size 16 needle on the one needle operations and a size 20 needle on the union special closer. The inside and outside quarters shall be closed seamed at the front and back on a union special closer, with a seam allowance of 3/32 inch. Then using a one needle machine, the side slashes shall be closed and the bootie bottom stitched to the quarters. The bootie shall then be seam sealed with the tape in 3.3.4.1. The second tape, as in 3.3.4.2, shall be applied to the heel area, and extended up the back seam above the counter area. The bootie shall not show any leakage when tested for waterproofness as described in 4.4.1.

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3.7.6 Upper leather fitting. Line marking patterns shall be used for all upper fitting. Inserts in cutting dies for stitch lines impression or die marking will not be allowed, except for the tongue and eyelet stay which will have stabs. Needles used for all fitting shall be no larger than size 18 except for Puritan vamping and automatic bartacking. All upper stitching shall be 8 to 10 stitches per inch, except Puritan vamping.

3.7.6.1 Vamping quarters to vamp. Vamping shall be done on a 3-N Puritan machine with three rows of stitching, 3/32 inch from the vamp top-line edge. Spacing between the three rows shall be 3/32. This upper stitching shall have 6 stitches per inch.

3.7.6.2 Eyelet stay to quarter. The eyelet stay shall be stitched to the upper with a single row stitch 3/32 from the back edge of the eyelet stay.

3.7.6.3 Closing back of upper. Quarters and vamp shall be closed at the back using overcasting stitch. The back seam shall be closed with the stitching closely positioned to the edge of the quarters and vamp on the grain side. The quarters and vamp shall then be rubbed open so that the quarters and vamp butt rather than lap.

3.7.6.4 Backstay fitting. The backstay shall be stitched to the upper with two rows stitching 3/32 from outside edge of backstay. There shall be 3/32 space between the two rows of stitching. Excess backstay shall be trimmed off at top edge of backstay even with quarter.

3.7.6.5 Thermal insulation to quarter. An adhesive, specified in 3.3.17, shall be applied to the quarter insulation and then the quarter insulation shall be placed on the flesh side of the quarters.

3.7.6.6 Gusset tongue fitting. The gusset tongue shall be stitched to the vamp grain side out, with two rows of stitching, one row not more than 3/32 inch from the edge of the tongue, and a second row placed not more than 1/4 inch from the edge of the tongue. The tongue shall be stitched to the quarter eyelet stay at top edge aligning nick-ins. The tongue shall then be stitched to the quarter eyerow with a stitch 3/32 from outside edge and a row 3/32 from existing eyerow stitch and trimmed flush with 1/8 inch trim.

3.7.6.7 Boxstitch eyelet stay. The eyelet stay shall be stitched to vamp at bottom of eyelet stay. The stitch shall start in the middle at the bottom of eyelet stay, 3/32 from edge, up the sides of eyelet stay to marker line and across, down eyelet stay and across bottom of eyelet stay interlocking stitches at bottom. This area shall then be pounded.

3.7.6.8 Bootie lining to upper fitting. The lining shall be placed inside the upper and stitched flush to the topline of the upper. The ends shall be placed under the gusset tongue and a bar tack will be added here for reinforcement. The lining shall then be tacked to front center of tongue, then stitched across front of tongue along marker line. Lining shall be trimmed at top across tongue.

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3.7.6.9 Foam padding and collar fitting. The collar shall be stitched flesh out even with the top edge of the quarter with one row of stitching 3/32 inch from edge. The collar foam shall be cemented with an adhesive specified in 3.3.17, flush to the top edge of the lining, placed so that it extends to the inside eyerow stitching on each side of the gusset. The collar shall be turned over the padding and stitched with one row not more than 3/32 from the edge and this stitching shall be extended to secure the end of the collar 3/32 inch from the edge of the eyerow and end with a three-stitch back tack at the eyerow edge at both ends of the collar. Excess collar leather shall be cut off flush to the eyerow edges.

3.7.6.10 Eyeletting. The eyelets shall be equally spaced on each quarter. All sizes shall have two eyelets on each quarter.

3.7.6.11 Attaching speedlacer assembly. The speed lace loops shall be equally spaced on each quarter. All sizes shall have six speedlace loops on each quarter. The speed lace loops shall be placed perpendicular to the edge of the quarters and parallel to each other, and shall be securely riveted.

3.7.6.12 Lacing for lasting. The quarters shall be laced by hand or by machine in the first eyelet with a lace opening 7/8 inch.

3.7.6.13 Automatic bartacking. The bartack shall be done by the automatic method with tying of locking ends. When an automatic stitching machine is used a size 20 needle may be used. There shall be two bartacks on each eyelet stay. The first bartack shall be below and parallel with the bottom topstitch line and the second shall be perpendicular with the outside box stitch line.

3.7.7 Insole assembly. The stuck on rib is applied to the insole on the printed side, then the plain side is cemented and allowed to dry. The counter, which has been coated on the inside with a cement in 3.3.17 and fitted in accordance with table V shall then be stapled to the rib edge of the insole. There shall be five to seven staples to each counter. A cloth binding shall be stitched to the top edge of each of the counters before assembly.

3.7.8 Lasting. The correct size and widths of uppers, lasts, insoles and box toes shall be assembled. The upper/bootie shall be slipped on the last with the upper pulled up and away from the heel area. Nylon tape shall then be placed on the toe area for abrasion protection. Cement shall then be applied to the toe and heel area and the assembled insole then placed to the bootie and pressed to bond insole to bootie. Cement shall be put on the heel area and the upper pulled into place at the heel area. The preformed plastic toe, which has been coated on the inside with a cement, shall then be put in place. A last slip material shall then be applied to the outside of the box toe for ease in pulling the leather during toe lasting. Uppers may be conditioned by any suitable means at toe lasting. Uppers shall be drawn over the last with proper tension on the pulling-over machine, cementing shall be done in conjunction with toe lasting, so that the toe is smoothly wiped-in

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and the toe staples attached securely around the toe at the base of the insole rib and the boxtoe. The sides shall then be pulled into place with hand pliers and the staples put around the shank and heel area through the base of the insole rib, counter, and upper. Excess upper shall then be trimmed at bottom to prepare for inseaming stitch.

3.7.9 Inseaming. Inseaming shall be done by machine with the thread waxed and with proper tension on the thread, using a needle not larger than number 41, and with not less than 3-1/4 stitches to the inch. The welting shall be inseam stitched to the bottom of insole rib from rib end to rib end and shall catch counter, box toe, and upper.

3.7.9.1 Inseam trimming. Excess upper, counter, and box toe shall be trimmed to the top of insole rib.

3.7.10 Shank fittings.

3.7.10.1 Shank fitting-bottom filling. The shanks shall be selected for applicable size in accordance with the shank fitting schedule specified in 3.3.11.1. The shanks shall be inserted in position with the shank board cemented to the shank, filling the cavity between the edges of the vamp from the ball line rearward to the back of the heel seat. The forward end of the shank shall be flush with the insole and shall fit the contour of the shoe bottom back of the ball line. The bottom filler shall be firmly pressed into the insole channel from the toe, extending to the forward end of the shank with a uniform even surface. The bottom shall present a flat smooth surface for sole laying.

3.7.10.2 Shank fitting - bottom filling (alternate). An alternate method of bottom filling shall be to insert the shank into position and then fill the entire insole channel from toe to heel with bottom filler, thus eliminating the need for a shankboard. The bottom filler shall be firmly pressed into the insole channel to form a uniform even surface. The bottom shall present a flat smooth surface for sole laying.

3.7.11 Midsole assembly. A coating of cement shall be applied to both the welt and midsole and allowed ample time to dry. The midsole shall be activated and spotted to the welt and then pressed on a sole press with sufficient pressure to ensure a good bond. The midsole shall then be rough rounded on a rough rounding machine.

3.7.12 Welting to midsole stitch. The welting and midsoles shall be stitched together on a lockstitch machine using waxed polyester thread using a number 47 needle and awl.

3.7.13 Sole laying. The sole shall be washed to remove any mold release and/or dirt from surface, then roughed and blown free of dust. The toe area of the outsole shall then be primed with an adhesive specified in 3.3.17, then cemented over the entire surface. The midsole shall be wire brushed, then primed and cemented over entire area. Both parts shall then be force

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dryed at 200°F for 4 minutes, then allowed to air dry for 20 to 30 minutes. They shall be placed on a flash activator at 125°F, then the outsole shall be placed to the midsole and be put on a sole laying machine with pressure. The soles shall be of adequate size and shall be laid evenly to allow for adequate sole extension. The outside edges shall then be pounded to secure the midsole to the outsole.

3.7.14 Rough rounding. Soles shall be uniformly rounded by any suitable means for the specified edge extension of sole and welt as specified in 3.7.17.

3.7.15 Sole toe stitching. Soles and welts shall be stitched together on a lockstitch machine using waxed polyester thread using a number 47 needle and awl. The stitch shall extend only around the toe area.

3.7.16 Pulling lasts. The lasting laces shall be cut from the uppers and lasts removed.

3.7.17 Edge trimming. Edges shall be trimmed using cutter #3x12-538KI-260 (12 tooth) or equal (see 6.7) to produce a rounded edge allowing for the required finished extension to be 3/16 ± 1/16 inch.

3.7.18 Sole finishing. Sole edge shall be trimmed square and smoothly scoured then stained to match the sole bottom.

3.7.19 Upper finishing.

3.7.19.1 Preparation. The boots shall be cleaned, removing accumulated dirt, wax, cement, and other foreign matter. All thread ends shall be trimmed.

3.7.19.2 Inserting molded cushion insole. An insole as specified in 3.3.20 shall be inserted in each boot.

3.7.20 Final finish. The boots shall be repaired and filled. All raw edges, including vamp line, backstay and front edges shall be stained to match upper leather. An overcoat of a bright spray shall then be applied. Materials and methods shall be as specified in 3.3.18 through 3.3.18.3.

3.7.21 Pairing and lacing. The boots shall be paired. A lace shall be put through the bottom eyelets of each boot, with the lace ends placed inside the boot.

3.7.22 Repairs of stitching.

3.7.22.1 Upper fitting stitches. Repairs of types 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end stitching.

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b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item at fitting room release, shall be repaired by overstitching. The stitching shall start one stitch in back of the defective area and continue one stitch beyond the defective area onto the existing. Repair stitching may not be done in a way that the waterproof bootie lining will be penetrated.

3.7.22.2 Inseaming and sole stitching. Repairs of inseaming and outsole stitches shall be in accordance with 3.7.22.1, a. and b., and in addition thread cuts occurring as a result of trimming operations shall be repaired in the same manner as specified for thread breaks.

3.8 Leakage. The finished boots shall be tested for leakage as specified in 4.4.4.

3.9 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects shall be removed from production and replaced with nondefective and properly matched components.

3.10 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and tested for the characteristics specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed in table X conform to the specified requirements.

TABLE X. Component tests

Material	Characteristic	Paragraph	Test Method
Insoles	Material identification	3.3.2	-
Laminated cloth	Weight (oz/sq yd)	3.3.3.3	5041 1/
	Thickness (inches)	3.3.3.3	5030 1/ 2/
	Thermal shrinkage machine (%):		
	Sample mean	3.3.3.3	4.5.6
	Lot mean	3.3.3.3	4.5.6
	Thermal shrinkage transverse (%):		
	Sample mean	3.3.3.3	4.5.6
	Lot mean	3.3.3.3	4.5.6
	Moisture vapor transmission rate ($g/m^2/24$ hr):		
	Procedure B	3.3.3.3	E 96 3/ 4/
	Procedure BW	3.3.3.3	E 96 3/ 5/

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TABLE X. Component tests (cont'd)

Material	Characteristic	Paragraph	Test Method
Laminated cloth (cont'd)	Water permeability:		
	Initial 3 psi, 5 minutes	no leakage	6/
	After aircraft fluid	no leakage	7/
	Colorfastness to:		
	Water	3.3.3.3.3	AATCC 107
	Perspiration	3.3.3.3.3	AATCC 15
	Crocking	3.3.3.3.3	AATCC 8
Laminated tape	Material identification	3.3.4.1	-
Film tape	Material identification	3.3.4.2	-
Collar foam	Material identification	3.3.5	-
Thermal insulation	Material identification	3.3.6	-
	Thickness	3.3.6	2/
Ribbing fabric	Material identification	3.3.7.1	-
	Shear strength	3.3.7.2	4.5.4
	Stitch strength	3.3.7.2	4.5.4
Box toe	Material identification	3.3.8	-
	Thickness	3.3.8	-
Shank fiberglass	Material identification	3.3.11	-
	Width	3.3.11	-
Staples	Material identification	3.3.12.1	-
Eyelets	Material identification	3.3.12.2	-
Speed lace loops	Material identification	3.3.12.3	-
	Finish	3.3.12.3	-
Bottom filler	Material identification	3.3.13.1	-
Outersole	Abrasive index	3.3.14.3	D 1630 3/
	Hardness	3.3.14.3	D 2240 3/

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TABLE X. Component tests (cont'd)

Material	Characteristic	Paragraph	Test Method
Outersole (cont'd)	Volume swell	3.3.14.3	6001 and 6211 8/
	Trouser tear	3.3.14.3	FIA 325 9/
	Nail tear	3.3.14.3	FIA 320 9/
	C tear	3.3.14.3	D 624 3/
	Ross flex	3.3.14.3	D 1565 3/
Adhesive	Material identification	3.3.17	-
Finishing materials	Compatability	3.3.18	-
Insole, cushion	Material identification	3.3.20	-
	Thickness	3.3.20	-
	Density	3.3.20	D 3574 (Test A) 3/
	Tensile strength	3.3.20	D 3574 (Test E) 3/
	Compression set	3.3.20	D 3574 (Test D) 3/
	Compression set after hydrolysis	3.3.20	D 3574 (Test D) 3/
Bootie	Waterproofness	3.7.5	4.5.1

1/ Refers to FED-STD-191.

2/ Specimen size: 3.5 inch diameter, total load; 0.25 psi, loading time: 5 seconds.

3/ Refers to ASTM Standards.

4/ The back side of the laminate cloth shall face the water and the free stream air velocity shall be 550 ± 50 FPM as measured 2 inches above the fabric specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall be run for 24 hours and weight measurements shall be taken at only the start and completion of the test. At the start of the 24 hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 ± 1/16 inch. Five specimens shall be tested. The test chamber shall be 73.5° ± 1°F and relative humidity shall be 50 ± 2 percent.

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5/ The back side of the laminate cloth shall face the water, the free stream air velocity shall be 550 ± 50 FPM as measured 2 inches below the fabric specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall be run for 2 hours and weight measurements shall be taken at only the start and completion of the test. Five specimens shall be tested. The specimen shall be sealed in any manner which prevents wicking and/or leaking of water out of the cup. The test chamber shall be 73.5° ± 1°F and relative humidity shall be 50 ± 2 percent.

6/ The water permeability shall be measured as specified in Method 5516 of FED-STD-191, except that the back side of the laminated cloth shall contact the water. The hydrostatic head shall be 50 centimeters and shall be held for five minutes. Leakage is defined as the appearance of water any place within the 4-1/2 inch diameter test area. The test may be performed using any device which tests the same specimen area at the equivalent pressure. In cases of dispute, the apparatus described in Method 5516 of FED-STD-191 shall be used.

7/ One specimen per sample unit shall be tested for leakage after exposure to aircraft fluids. The specimens shall not be less than 6 inches in diameter. Specimens shall be placed on a flat surface and 1 mL of JP-4 (jet fuel conforming to MIL-T-5624) spread over the middle of the back side, followed by 1 mL of de-icing fluid conforming to MIL-A-8243. Place the horizontal specimens flat in an air circulating over at 50°C for 30 minutes. Remove from oven and test immediately as specified in footnote 6/.

8/ Refers to FED-STD-601.

9/ Refers to Footwear Industries of America Standards.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process examination. The defects found during in-process examination shall be classified in accordance with 4.4.2.1.1, 4.4.2.1.2 and 4.4.2.1.3. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk in tables XI through XIV shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

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4.4.2.1.1 Pre-fitting examination of cut parts. The cut parts shall be examined for the defects listed in table XI before fitting. The lot size shall be expressed in units of cut parts. The sample unit shall be one cut part. Defects shall be major when seriously affecting serviceability or appearance, and minor when affecting serviceability or appearance, but not seriously.

TABLE XI. Cut parts defects

Examine	Defect
Upper leather	Not full grain
	Leather deeply snuffed; i.e., fiber structure damaged
	Any grub or tick damage, scratches, brands, or boney leather
	Fat wrinkles or veins
	Stretchy vamp
	Leather embossed or printed
Gusset-tongue and collar leather	Flanky
	Loose flesh, boney, or boardy
Quality of leather	Thickness exceeding the maximum or less than the minimum specified
	Rough fiber on flesh side
	Off-stretch cut
	Slaughter cut

4.4.2.1.2 Examination of uppers after all fitting. The upper assemblies shall be examined for the defects listed in table XII. The lot size shall be expressed in units of upper assemblies prepared for lasting. The sample unit shall be one completely fabricated upper assembly prepared for lasting.

TABLE XII. Uppers after all fitting defects

Examine	Defect	Classification Major	Minor
Construction and workmanship (general)	Any component missing or not specified type	101	
	Any component misplaced or not affixed as specified	*	
	Any cut, tear, hole, repair, or factory damage	*	
	Backseam not rubbed down		201

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TABLE XII. Uppers after all fitting defects (cont'd)

Examine	Defect	Classification Major	Minor
Bootie lining	Torn, loose, or wrinkled		*
	Not properly mated with upper		*
Quality of leather	Thickness more than 1/2 ounce less than minimum specified	102	
	Thickness less than specified minimum but not extending 1/2 ounce less		202
	Thickness more than the maximum specified		203
	Off-stretch cut	103	
	Slaughter cut		*

4.4.2.1.3 In-process examination of boot before bottom filling. The partially fabricated boots shall be examined for the defects listed in table XIII. The lot size shall be expressed in units of partially fabricated boots. The sample unit shall be one partially constructed boot assembled to the point after lasting and attachment of shank, but before the application of the bottom filler.

TABLE XIII. Boot before bottom filling defects

Examine	Defect	Classification Major	Minor
Bottom of boot	Any component missing or not specified type	101	
	Shank not properly positioned or not fitting contour of boot bottom	102	
	Shank wrong size, malformed, or not securely attached	103	
	Upper damaged		*
	Poor heel seat, side, or toe lasting		*
	Inseam not properly trimmed		201
	Inseam stitches broken, two or more consecutive skipped inseam stitches, or inseam stitches cut or damaged during trimming operation - not repaired	104	
	- not repaired as specified		202

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TABLE XIII. Boot before bottom filling defects (cont'd)

Examine	Defect	Classification Major	Minor
Bottom of boot (cont'd)	Inseam stitches not in insole rib		203
	End of counter not caught in inseam stitching	*	
Upper part of boot	Upper not tightly pulled down to last	*	
	Quarters not laced as specified		204
	Upper in lasted area not sufficiently roughed or cleaned after roughing	105	
	Excessive roughing or scouring	*	

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table XIV. The lot size shall be expressed in units of boots. The sample unit shall be one completely fabricated boot and the selection shall be by pairs. The inspector shall check to see that insoles are inserted in the boots. For the pairing examination, and when determining possible differences in outersole thickness, the pair shall be examined together. Defects of pairing shall be classified as a single defect. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (combined major and minor) defects.

TABLE XIV. End item visual defects

Examine	Defect	Classification Major	Minor
Pairing	Not properly mated; i.e., not right and left of same size	101	
	Variation in color, luster, or appearance	*	
	Variation of more than 1/4 inch in height of pair	102	
	Box toe crooked, long or short	*	
	Difference in outersole thickness between left and right boots:		
	- more than 3/32 inch	103	
	- more than 1/16 inch but not more than 3/32 inch		201

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TABLE XIV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Cleanness	Any nonremovable spot, stain, or foreign matter affecting appearance		202
Color and finish	Not specified color	104	
	Finish streaky, chipped, or flaky on upper		203
	Any raw edges not stained to match upper leather		204
Design, type and style	Not as specified	105	
Construction and workmanship (general)	Any cut, tear, hole, repair, or factory damage	*	
	Any component or assembly omitted or misplaced, operation omitted, or not properly performed (unless otherwise classified herein)	*	
Seams and stitching (upper)	Open seam not repaired	106	
	Repair of open seam not as specified		205
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When the above conditions occur on the inside of the boot it shall be scored as a * defect.		
	Tight tension resulting in puckering or cutting of the leather	*	
	Loose tension resulting in a loosely secured seam	*	
	Wrong stitch type	107	
	Stitching omitted where required	*	
	Gage of stitching not as specified or irregular	*	
	Thread ends not trimmed throughout the boot		206
	Needle holes or needle chew	*	

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TABLE XIV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Vamp stitching at outersole junction	One or more rows of defective stitching not repaired	108	
	Repaired but first stitch more than 1/4 inch from outersole junction	109	
	Repaired with one stitch in lieu of two or more stitches		207
Outersole and heel	Thickness not as specified	110	
	Edge scouring is irregular, affecting appearance		208
	Blisters:		
	- 1/2 inch or more in longest dimension	111	
	- penetrating 1/3 or more the thickness of the rubber	112	
	- more than three blisters, more than 1/8 inch but less than 1/2 inch in longest dimension	113	
	- one, two or three blisters, more than 1/8 inch but less than 1/2 inch in longest dimension		209
	Any other defect in tread or edge area	*	
	Incomplete or incorrect vulcanization of outersole or heel to uppers (i.e., rubber not securely attached)	*	
Eyelets	Number of eyelets not as specified but each row has the same number		210
	Not the same number of eyelets in each row	114	
	Eyelets not properly spaced within the row of misalignment between the rows to an extent interfering with proper lacing	*	
	Edge of any eyelet less than 1/16 inch from edge of quarter	115	
	Any eyelet not securely clinched		211

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TABLE XIV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Speed laces	Number of speed laces loops not as specified but each row has the same number		212
	Not the same number of loops in each row	116	
	Loops not properly spaced within the row of misalignment between the rows to an extent interfering with proper lacing	*	
	Loops not perpendicular to edge of quarters or parallel to each other		213
	Any loop not securely riveted	117	
	Four splits in rivet clinch roll		214
	Five or more splits in rivet clinch roll	118	
	Distance from center of rivet to edge of quarter less than 7/32 or more than 9/32 inch	*	
Marking and instruction tag	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size		215
Bootie lining	Torn, loose or wrinkled	*	
	Not properly mated with upper		216
Cushion insole	Missing, wrong size, or wrong side up	119	

4.4.4 End item testing. The finished boot shall be tested for the characteristics listed in table XV. The sample unit for all tests shall be one boot. All test results shall contain the individual values utilized in expressing the final results. For all tests except leakage, the sample size shall be five boots regardless of lot size and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified. For the leakage test, the sample size shall be 10 percent of the lot regardless of lot size and the lot shall be unacceptable if two or more sample units fail the leakage test.

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TABLE XV. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as Pass or fail	Results reported as Numerically to nearest
Bond strength (outersole)	3.3.14.4	4.5.2	1	-	1 lb.
Leakage	3.8	4.5.3	1	X	-

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall 2.5.

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling
Content	Numer per container is more or less than required

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified

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Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Bootie lining tests. The bootie will be tested for waterproofness before it is inserted into the boot using the test apparatus listed in 6.5.13. The test will be performed as follows:

a. Inflate the bootie with air to 1.0 psi gage static air pressure (measured inside the bottie) and clamp the top of the bootie to seal from air leakage.

b. Submerge inflated bootie in a water bath such that the toe of the bootie is at least 5 inches below the water's surface for 2 seconds. Remove from bath and wipe entire bootie surface with hand.

c. Re-submerge inflated bootie to the same position for at least 10 seconds.

d. While submerged, the bootie shall be inspected for air leakage. The bootie fails if, at any single location on the exterior of the bootie, there is a discreet air bubble more than once every 7 seconds.

e. No more than two repairs shall be made on each bootie. All repairs shall be made with the seam sealing tape referenced in 3.3.4.1.

4.5.2 Bond strength test of outersole to midsole.

4.5.2.1 Specimen. The specimen shall be a combined midsole-outsole unit which has aged at least 2 days. The outsole shall be separated from the midsole for a distance of approximately 2-1/2 inches from the toe end of the specimen.

4.5.2.2 Apparatus. A power-driven portable gage tong adhesion machine or an approved portable testing device of equal performance shall be used. The rate of travel of the power actuated grip shall be 2 inches per minute. The machine shall be operated without any device for maintaining maximum load indication.

4.5.2.3 Procedure. The separate toe ends of the specimen shall be clamped in the jaws of the machine, with the jaws 2 inches apart. The specimen shall extend outward at right angles to the direction of the application load. The machine shall be started and the outsole and midsole shall be pulled apart for a distance of 1 inch. At that instant, the load indicated on the machine shall be read and recorded. The load shall be divided by the width of the specimen at the corresponding line of separation.

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4.5.3 Boot leakage test. Remove the ventilating or cushioned insert. Place the boot on a horizontal plane. Fill the boot with about 4 inches of water. Any boot displaying leakage after 2 hours shall be a test failure. The boots tested for leakage shall be thoroughly dried.

4.5.4 Testing of rib strength. The rib shall be tested for compliance with the requirements of 3.3.7.2. Method 2061 of FED-STD-311 shall be used for shear strength test, and Method 2171 of FED-STD-311 shall be used for stitch tear strength. The sample unit shall be two ribbed insoles, one for each test, and the sample size shall be 5, regardless of the lot quantity. Requirements are applicable to the sample unit for individual unit requirements and lot average for average requirements. The lot shall be unacceptable if one or more sample units or the lot average of all units fail to meet the requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

4.5.5 Thermal shrinkage resistance test. Three 15 inch by 15 inch specimens shall be cut from each sample unit. All specimens to be tested shall be conditioned in accordance with FED-STD-191 (Section 4, Atmospheric Conditions for Testing). Specimens shall be tested not more than 5 minutes after removal from conditioning. Specimen marking and measurements shall be in accordance with the procedure specified in AATCC Test Method 135. The forced circulating air oven shall achieve and maintain an air temperature of 181° $+5^{\circ}$, -0° for a period of not less than 10 minutes. Oven recovery time after the door is closed shall not exceed 1 minute. The specimens shall be mounted on a suitable rack such that the specimens are suspended by two metal hooks near each top corner with 12 $\pm$ 5 inches separation between hooks. The entire specimen shall be exposed to the circulating air so that it is not less than 2 inches from any oven surface or other specimen, and the airflow is parallel to the plane of the material. The rack assembly shall be placed in the oven for 10 minutes. Specimen exposure time shall begin when the oven has recovered to an air temperature of 181° $+5^{\circ}$, -0°F. The specimens shall be conditioned prior to shrinkage measurement.

4.5.6 Non-marking test for outersoles and heels. The edge of the outersole and heel shall be drawn across a sheet of white bond paper using moderate pressure. Any resulting marking which cannot be easily removed by rubbing with the fingers shall be considered a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation.

5.1.1.1 Unit pack. Each pair of boots shall be unit packed in a one-piece die-cut fiberboard set-up box. The boots shall be separated from each other by tissue paper conforming to type I of A-A-1249. The box shall be constructed from domestic corrugated single-wall minimum grade 200 fiberboard

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in accordance with PPP-F-320. Approximate inside dimensions shall be as specified in table XVI.

5.1.1.2 Alternate unit pack. When specified (see 6.2), each pair of properly mated boots shall be tied together by the ends of laces which have been inserted through the top speed lace of the outside quarter of each boot.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Six pack. Six pairs of boots of one size and width only, unit packed as specified in 5.1.1.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V2s of PPP-B-636. Unit packs shall be arranged one across the length, two across the width, and three in depth. The approximate inside dimensions of the unit pack and shipping container shall be as specified in table XVI. Each container shall be closed in accordance with Method III, waterproofed in accordance with Method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping except for type II, class F loads.

TABLE XVI. Unit pack and shipping container dimensions

Range of boot sizes	Unit pack dimensions	Shipping container dimensions	Pallet number
3 through 5	10-1/2 x 9-1/2 x 5	19-1/2 x 11 x 15-1/2	102
5-1/2 through 8	11-1/2 x 10 x 5	20-1/2 x 12 x 15-1/2	102
8-1/2 through 11	12-1/2 x 11 x 5-1/2	23 x 13-1/2 x 15-3/8	91
11-1/2 through 14	14 x 11-1/2 x 5-1/2	24 x 14-3/8 x 15-3/8	3

5.2.1.2 Alternate six pack. When specified (see 6.2), six pairs of boots of one size and width only, preserved as specified in 5.1.1.2, shall be arranged as shown in table XVII and packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table XVII. A sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with Method III, waterproofed in

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accordance with Method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

TABLE XVII. Alternate six pack dimensions and packing arrangements

(A) Approximate inside dimensions of shipping container. 1/

Range of boot sizes	Length, inches	Width, inches	Depth, inches	Pallet number
3 through 5	23-1/2	12	11	91
5-1/2 through 8	24-1/2	13	12	91
8-1/2 through 11	25-1/2	13-3/4	13	91
11-1/2 through 14	27	14-1/2	14	6

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5XN and above 12XW, the contractor shall determine the size container to use. The dimensions of the container shall be such that it will result in a snug fit of the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversize container will not be permitted.

(B) Packing arrangements. Arrangement of boots within the shipping container shall be in three layers with two pairs per layer. Each pair shall be in a flat position, laces to laces, with soles in opposing directions. The first two pairs of boots shall be placed end-to-end. The third pair shall be placed on top of the second pair also end-to-end. The fourth pair shall be placed on top of the first pair in the same manner as the third pair was positioned to the second pair. Each layer of boots shall be alternated in opposite direction as the previous layer.

5.2.1.3 Alternate twelve pack. When specified (see 6.2), twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as shown in table XVIII(B) and packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table XVIII(A). A sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container.

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Each container shall be closed in accordance with Method III, waterproofed in accordance with Method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to non-metallic strapping, except for type II, class F loads.

TABLE XVIII. Twelve pack shipping container dimensions and packing arrangements

(A) Approximate inside dimensions of shipping container. 1/

Range of boot sizes	Length, inches	Width, inches	Depth, inches	Pallet number
3 through 7-1/2	30	14-1/2	11-1/2	94
8 through 11-1/2	31	15-1/2	13-1/2	94
12 through 14	34	17	15	1

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5XN and above 12XW, the contractor shall determine the size container to use. The dimensions of the container shall be such that it will result in a snug fit of the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversize container will not be permitted.

(B) Packing arrangements. Arrangement of boots within the shipping container shall be in three layers with four pairs per layer. Each pair shall be in a flat position, laces to laces, with soles in opposing directions. The first two pairs of boots shall be placed end-to-end in the length of the container. The third pair shall be placed on top of the second pair with the heels next to the toe of the second pair. The fourth pair shall be placed on top of the first pair in the same manner as the third pair was positioned to the second pair. The first four pairs of boots constitute one layer. Each layer of boots shall be alternated in opposite direction as the previous layer.

5.2.2 Level B packing.

5.2.2.1 Six pack. Six pairs of boots of one size and width only, preserved as specified in 5.1.1.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Unit packs shall be arranged one along the length, two across the width and three in depth. The approximate inside dimensions of the shipping container shall be as specified in table XVI.

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Each container shall be closed in accordance with Method II as specified in the appendix of PPP-B-636.

5.2.2.2 Alternate six pack. When specified (see 6.2), six pairs of boots of one size and width only, preserved as specified in 5.1.1.2, shall be arranged as shown in table XVII(B) and packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table XVII(A). A sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with Method II as specified in the appendix of PPP-B-636.

5.2.2.3 Alternate twelve pack. When specified (see 6.2), twelve pairs of boots of one size and width only, preserved as specified in 5.1.1.2, shall be arranged as shown in table XVIII(B) and packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636, except that the abutting ends of the liner do not need to be joined. The approximate inside dimensions of the shipping container shall be as specified in table XVIII(A). A sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203 shall be placed between each layer of boots in order to protect the leather from damage by the speed lace loops. A sheet of this paper shall also completely cover the top and bottom of the contents of each container. Each container shall be closed in accordance with Method II as specified in the appendix of PPP-B-636.

5.2.2.4 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

5.3 Palletization. When specified (see 6.2), boots packed as specified in 5.2.2 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147 except that the use of a sheet of paper or fiberboard between every other layer is not required. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III or IV of MIL-STD-731. Each prepared load shall be bonded with primary or secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern number shall be as specified in the applicable table.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boots are intended for use by military personnel of the Department of Defense.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Size and width required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. When an alternate unit pack is required (see 5.1.1.2).
f. Levels of packing (see 5.2).
g. Type and class of unit load required (see 5.2.1.1).
h. When an alternate six pack is required for level A packing (see 5.2.1.2).
i. When an alternate twelve pack is required for level A packing (see 5.2.1.3).
j. When an alternate six pack is required for level B packing (see 5.2.2.2).
k. When an alternate twelve pack is required for level B packing (see 5.2.2.3).
l. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.4).
m. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal or as directed by the contracting activity.

6.5 Sources of supply.

6.5.1 Leather.

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6.5.1.1 Style Thule J74000. A suggested source for the Style Thule J74000 leather specified in 3.3.1.1 is Volunteer Leather, Nashville, TN 37202.

6.5.1.2 Style Boulder HH 13339. A suggested source for the Boulder HH 13339 leather specified in 3.3.1.1 is S.B. Foot, Red Wing, MN 55066.

6.5.2 Insoles. A suggested source for the Texorist insole specified in 3.3.2 is USM Corporation, Texon Division, 1190 Huntington Road, P.O. Box 365, Russel, MA 01071.

6.5.3 Bootie lining material. A suggested source for the bootie lining material specified in 3.3.3.1 is W.L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

6.5.4 Thermal quarter insulation. A suggested source for the thermal quarter insulation specified in 3.3.3.1.1.1 (for product designated as Thinsulate B400) is Thermal Insulation Products, 3M Center Building, 200-3E-04, St. Paul, MN 55144.

6.5.5 Seam sealing tape.

6.5.5.1 Seam sealing tape for bootie. A suggested source for the tape specified in 3.3.4.1 is W.L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

6.5.5.2 Seam sealing tape for heel and counter. A suggested source for the tape specified in 3.3.4.2 is Bemis Associates, Inc., One Bemis Way, Shirley, MA 01464-2527.

6.5.5.3 Seam sealing tape machine. A suggested source for the machine to apply the seam sealing tapes specified in 3.3.4.1 and 3.3.4.2 is W.L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

6.5.6 Box toes. A suggested source for the box toes specified in 3.3.8 is Brockton Sole and Plastic, 53 Spark Street, P.O. Box 2128, Brockton, MA 02403.

6.5.7 Fiberglass shank. A suggested source for the fiberglass shank specified in 3.3.11 is American Shoe Machinery Co., (designated "TRU-FIT"), 199 Newbury Street, P.O. Box 390, Danvers, MA 01923.

6.5.8 Shankboard. A suggested source for the Bonfoam 300HD (see 3.3.11.2) is Georgia Fibers, Inc., Buena Vista, VA 24416.

6.5.9 Outersole. A suggested source for the outersole specified in 3.3.14.2 is Quabaug Corp., (designated "VIBRAM"), North Brookfield, MA 01535.

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6.5.10 Molded cushion insoles. Suggested sources for the cushion insoles specified in 3.3.20 are Gemini Laminating Corporation (Product No. 139300), 35 Balch Street, Building K, P.O. Box 398, Beverly, MA 01915, Frelonic (Product No. FI-9503), 63 Grove Street, P.O. Box 169, Salem, MA 01970 or Rogers Foam, 145 Central Street, Somerville, MA 02145.

6.5.11 Microcellular foam. A suggested source for the microcellular foam meeting the requirements of 3.3.20 is Rogers Foam, Poron Division, East Woodstock, CT 06244.

6.5.12 Lasts.

6.5.12.1 M2673B-1 lasts. A suggested source for the M2673B-1 lasts specified in 3.6 is Vulcan Corp.

6.5.12.2 701 lasts. A suggested source for the 701 lasts specified in 3.6 is Jones and Vining Corp.

6.5.13 Bootie lining testing apparatus. A suggested source for the testing apparatus needed for testing in 4.5.1 is W.L. Gore and Associates, 297 Blue Ball Road, Elkton, MD 21921.

6.6 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO STANAG 2333. When amendment, revision, or cancellation of this document is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.8 Subject term (key word) listing.

Footwear
Insulated
Leather, black
Liner
Waterproof

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - NU

Review activities:

Army - MD
Navy - MC
DLA - CT

Preparing activity:

Army - GL

(Project 8430-0406)

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INSTRUCTION TAG

BOOT, INTERMEDIATE, COLD/WET

FACTS

These Intermediate Cold/Wet Boots (ICWB) have been designed to protect your feet from cold injury and frostbite in areas where moisture and cold are critical factors, and where the mean monthly temperature ranges between 14°F and 68°F. The boots should not be worn where temperatures fall below +10°F. (NOTE: For temperature below +10°F, the Cold Weather, Insulated, Rubber, Black or White, Vapor Barrier Boots should be worn.)

If your feet begin to get cold, button up all your garments and exercise the entire body by swinging the arms, stamping the feet, and rapid movement of the toes.

TRENCH FOOT AND FROSTBITE are serious cold injuries which can cause painful and permanent disability. These injuries can be prevented by proper care of your feet. No boot will keep your feet from freezing in cold/wet conditions if you remain motionless for several hours. Wear the ICWB properly and DO NOT BECOME A COLD WEATHER CASUALTY.

INSTRUCTIONS FOR USE

1. Wear with: 1 pair socks, wool, cushion sole.

2. Sizes. The boots are supplied in whole and half sizes 3 through 14, and in widths N, R, W and XW. In some instances it may be necessary to select a slightly larger size than normally worn in order to allow for normal swelling of the feet.

3. Pull on boots, seating heel firmly in place, then lace. Boots should fit snugly but not tightly. There should be approximately 1/2 inch additional length at toe.

4. Carry extra socks. Change socks often, at least once daily. Remove boots and dry feel as often as possible. Neglect of feet can make you a casualty.

5. When your boots become soiled or dirty, simply brush or wipe them clean with a damp cloth. Let them dry naturally at room temperature (do not dry next to radiator). Waterproofing is not required, however, use of conditioners such as mink oil will help to maintain the natural condition of the leather and extend the service life of the boots.

FIGURE 1. Instruction tag

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-44426A	1992 June 26

3. DOCUMENT TITLE
Boots, Intermediate Cold/Wet

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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